

Valuation-Momentum Basket

A Nifty 50 stock basket engineered to beat the NIFTY50 index on a risk-adjusted basis — riding **momentum**, but only where it is **earnings-backed** and not **frothy**.

Universe **49 Nifty 50 names** · Backtest **Aug 2017 → Jul 2026 (~9y)** · Rebalance **Quarterly** · Gross CAGR **16.1%** vs 10.6% · Sharpe **1.01**

Design principle — momentum RANKS, valuation GATES. Momentum picks the winners; valuation and earnings-backing decide who is *eligible* to be picked. The book holds the top 15 survivors, inverse-volatility weighted.

1 Data pipeline

Two SQLite stores feed the strategy: `nifty500_hrp.db` (prices & benchmarks) and `equinext.db` (fundamentals). Everything downstream reads from these.



DATA	SOURCE	SCRIPT	STORE
Daily prices (10y)	Yahoo Finance	<code>load_nifty50_data.py</code>	<code>nifty500_hrp.db → ohlcv</code>
Benchmarks (NIFTY50/500)	Yahoo Finance	<code>load_nifty50_data.py</code>	<code>nifty500_hrp.db → broad_indices</code>
Fundamentals (10y annual)	screener.in (scraped)	<code>scrape_screener.py</code>	<code>equinext.db → valuation_series</code>
Readable CSV exports	derived from DBs	<code>export_tables.py</code>	<code>exports/*.csv</code>

Coverage. 49 Nifty 50 names (TATAMOTORS excluded — delisted on Yahoo post-demerger). Daily prices and derived multiples span **2016-07-04 → 2026-07-03**. 48/49 have full valuation multiples; SBILIFE failed to parse on screener (insurer page) and runs momentum-only.

Valuation multiples — per share, all in ₹

screener is INR-native (no currency issues), stepped at fiscal year-end + **75-day reporting lag** (a year's results are "known" only after they'd have been announced — no look-ahead), then forward-filled onto daily prices.

```
shares(yr) = Net Profit / EPS                                (both from screener P&L)
pe          = close / EPS
pb          = close / [(Equity Capital + Reserves) / shares]
ev_ebitda   = (close + Borrowings/shares) / (Operating Profit / shares)  [non-financials]
fcf_yield   = (Free Cash Flow / shares) / close                        [non-financials]
```

2 Selection criteria — the four-step funnel

Applied at every rebalance, using **only data on/before the rebalance date**.

1 Universe — eligibility

GATE

`ctx.universe(as_of)` — base list is Nifty 50 (49 names), then three liquidity/listing gates:

Liquidity	median daily traded value (close × volume) over trailing 63 days ≥ ₹5 crore
Free-float	free-float market cap ≥ ₹100 crore
History	≥ 200 days price history; valuation gates require ≥ 60 valuation rows (<code>MIN_VAL_OBS</code>)

2 Valuation froth gate

FILTER

Composite **own-history froth percentile** over a trailing 5-year window (`valuation_froth_percentile`), averaged across whatever multiples the name has:

MULTIPLE	MEANING	DIRECTION
P/E	price ÷ EPS	higher = richer
P/B	price ÷ book value/share	higher = richer
EV/EBITDA	enterprise value ÷ EBITDA	higher = richer
FCF yield	free cash flow ÷ market cap	higher = cheaper (inverted)

Financials (banks / NBFCs / insurers) use **P/E + P/B only**.

Rule: drop names with composite froth percentile > **0.80** (`FROTH_MAX`) — the own-history expensive extreme.

3 Momentum score

RANK

Cross-sectional composite z-score (each factor winsorized at 1%/99%, z-scored, equal-weighted — `composite`) of three signals:

SIGNAL	DEFINITION	PRIMITIVE
12-1 momentum	12-month return skipping the last month	<code>momentum_12_1</code>
Distance from 52-wk high	close ÷ trailing 52-week high	<code>distance_from_high</code>
Volume trend	21-day avg volume ÷ 126-day avg volume – 1	<code>volume_trend</code>

4 Earnings-backed gate

FILTER

`erating_decomposition` splits the trailing 12-month price return into earnings-driven vs multiple-driven components.

Rule: keep only names where earnings contribution ≥ multiple contribution — drops pure multiple-expansion "story stocks".

Selection & weighting

Combine	momentum ranks the survivors of Steps 2 & 4. If gates leave fewer than 8 names (<code>N_MIN</code>), the froth gate relaxes so the book isn't too thin.
Hold	top 15 by momentum (<code>N_HOLD</code>).
Weight	inverse-volatility — $1 \div (252\text{-day annualized vol})$, normalized to sum to 1. Calmer names get more.

3 Backtest mechanics

Shared harness `equinext/backtest.py` , runner `scripts/run_backtest.py` .

Period	2017-08 → 2026-07 (~9 years)	No-trade band	0 default (tested at 3%)
Rebalance	Quarterly (quarter-end)	Benchmark	NIFTY50 price index (^NSEI)
Trading cost	25 bps / side on traded fraction	Returns	daily; weights drift between rebalances
Tax (FIFO)	20% short-term (<1y), 12.5% long-term (>1y)	Look-ahead	signals ≤ rebalance date; fundamentals lagged 75d

Why quarterly? A 12-1 momentum signal barely changes month to month. Monthly rebalancing churned ~1041%/yr and — after costs and short-term-gains tax — dragged after-tax CAGR to ~5%. Quarterly cut turnover to ~337%/yr and lifted after-tax CAGR to ~12%. **Cadence was the single biggest performance lever.**

4 Results — 2017-08 → 2026-07, quarterly

CAGR (GROSS)

SHARPE

MAX DRAWDOWN

₹100 BECAME

16.1% NIFTY50 10.6%	1.01 NIFTY50 0.63	-32.0% NIFTY50 -38.4%	₹360 NIFTY50 ₹248
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METRIC	BASKET	NIFTY50
CAGR (gross)	16.1%	10.6%
CAGR (after tax)	12.3%	—
Sharpe	1.01	0.63
Max drawdown	-32.0%	-38.4%
₹100 became	₹360	₹248

Regime test (excess vs NIFTY50): 2018 +7.1% · 2020 (COVID) +14.7% · 2022 (rotation) +1.6% · COVID crash Feb–Mar '20 fell -31.3% vs index -36.5%. No momentum crash.

Ablation — do the valuation gates earn their place?

Same data, same period, gates OFF (momentum_only) vs ON (valuation_momentum):

METRIC	GATES OFF	GATES ON	GATE EFFECT
CAGR (gross)	13.3%	16.1%	+2.7%/yr
Sharpe	0.74	1.01	+0.27
Max drawdown	-37.0%	-32.0%	+5 pts
Turnover	394%/yr	337%/yr	lower

The gates **help in reversals** (2020 +11.3%, 2022 +9.1%) and **cost in bulls** (2021 -14.4%, 2023 -7.1%) — insurance that nets positive across a full cycle, improving risk metrics most. This is the documented value+momentum interaction (Asness, "Value and Momentum Everywhere").

5 Known limitations — honest caveats

- Survivorship bias** — universe is *today's* Nifty 50 applied backward; relegated names are excluded, inflating the CAGR edge. Fix: point-in-time index membership. Drawdown / crash-protection findings are far more robust to this than the CAGR.
- Thresholds not yet stress-tested** — froth 0.80, 15 holdings, momentum windows are sensible a-priori defaults, not swept. Robustness sweep still owed.
- Annual fundamentals granularity** — screener annual + 75-day lag; within a year multiples move only with price, not quarterly earnings.
- Price-return benchmark** — excludes dividends on both sides (fair comparison, but absolute CAGRs understate reality ~1–1.5%; TRI is the eventual benchmark).
- SBILIFE** runs momentum-only (screener parse failure).

6 How to run

```
# 1. Load data (one-time / refresh)
python scripts/load_nifty50_data.py           # 10y prices + benchmarks
python scripts/scrape_screener.py             # 10y fundamentals → valuation_series

# 2. Backtest
python scripts/run_backtest.py valuation_momentum --rebalance Q  # the strategy
python scripts/run_backtest.py momentum_only --rebalance Q      # gates-off ablation

# 3. Reports (close the .xlsx in Excel first)
python scripts/export_tables.py                # refresh readable CSVs
python scripts/format_monthly_report.py valuation_momentum      # formatted Excel
```

Optional flags: `--start YYYY-MM-DD` (custom window), `--band 0.03` (no-trade band).

7 File map

FILE	ROLE
baskets/valuation_momentum.py	the strategy — <code>ValuationMomentumBasket.select()</code> + <code>MomentumOnlyBasket</code> ablation
equinext/primitives.py	shared signals: momentum, distance-from-high, volume-trend, froth percentile, re-rating, vol
equinext/scoring.py	winsorize / z-score / composite
equinext/backtest.py	rebalance loop, costs, FIFO tax, cadence
equinext/universe.py	liquidity + free-float + listing gates
equinext/data/valuation.py	reads <code>valuation_series</code>
scripts/load_nifty50_data.py	Yahoo prices + benchmarks loader
scripts/scrape_screener.py	screener.in fundamentals scraper → 10y multiples
scripts/run_backtest.py	generalized runner (any basket + cadence)
scripts/export_tables.py	regenerate readable CSVs from live DB
scripts/format_monthly_report.py	formatted Excel report
exports/	readable CSV/XLSX snapshots